Numerical Suppression of Magnetized Richtmyer–Meshkov Instability in Hypervelocity Jet Stagnation

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(Dated: January 25, 2026)

The Richtmyer–Meshkov instability degrades stagnation yield in magneto-inertial fusion by driving turbulent mixing across shocked density interfaces. I present a two dimensional ideal MHD solver built from scratch around the five-wave HLLD Riemann scheme, MUSCL Van Leer reconstruction, and three-stage SSP Runge-Kutta time integration. Divergence-free magnetic fields are maintained via a coupled GLM-Powell cleaning strategy. The solver is verified against the Brio-Wu shock tube, a linear Alfvén wave convergence test recovering second-order spatial accuracy, and an isolated contact discontinuity. Applied to a Mach 10 shock impacting a perturbed density interface (density ratio 3:1), the simulations show that a transverse magnetic field of moderate strength ($B_y = 1.5$) suppresses the integral mixing width, collapses local enstrophy at the interface, and preserves interfacial coherence. Early time perturbation growth matches Richtmyer's impulsive linear model before magnetic stabilization dominates.

I. THE STAGNATION PROBLEM

In magneto-inertial fusion, a set of hypervelocity plasma jets converge on a magnetized target and compress it to thermonuclear conditions. The stagnation interface, where the imploding liner meets the fuel is a textbook breeding ground for the Richtmyer-Meshkov instability. A single reflected shock crossing a perturbed density jump is enough to seed baroclinic vorticity, roll up the interface into spikes and bubbles, and turbulently mix cold liner material into the hot fuel. RMI is the yield killer. It does not care how clever your driver is or how symmetric your implosion looks on paper; once the mixing zone is thick enough, the fuel cools below ignition threshold and the shot is dead.

My interest in this problem is practical. I spent most of my time at Swansea chasing high-Mach discontinuities through Godunov-type solvers, learning the hard way that if you cannot resolve a contact wave cleanly, everything downstream of it is noise. Commercial CFD packages hide the Riemann solver behind a GUI and call it a feature. I wanted the opposite, a code where every flux evaluation, slope limiter, and every divergence-cleaning term is something I wrote, tested, and understand at the register level. This paper documents that code and the physics it captures.

The central question is simple. A transverse magnetic field threads the pre-shock medium. After the shock crosses the density interface, magnetic tension $(\mathbf{B} \cdot \nabla)\mathbf{B}/\mu_0$ acts as a restoring force on the deforming interface. How strong does the field need to be to kill the instability before it transitions to turbulence?

II. NUMERICAL ARCHITECTURE

A. Governing Equations

I solve the two-dimensional ideal MHD system in conservative form,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = \mathbf{S}, \tag{1}$$

where the state vector is $\mathbf{U} = (\rho,\, \rho v_x,\, \rho v_y,\, \rho v_z,\, B_x,\, B_y,\, B_z,\, E,\, \psi,\, \rho C)^T$. The last two entries are the GLM scalar ψ and a passive color field $C \in [0, 1]$ that tracks material mixing. The total energy density closes the system through the equation of state for an ideal gas with $\gamma = 5/3$:

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2. \tag{2}$$

B. HLLD Riemann Solver

The choice of Riemann solver determines what physics you actually resolve. Standard HLL, which I have tested for this problem in the early stages , averages the interwave region into a single state, which smeared contact and Alfvén discontinuities beyond recognition. For MHD problems where the contact wave carries the density interface and the Alfvén waves carry magnetic tension information, this is unacceptable.

I implemented the HLLD approximate Riemann solver of Miyoshi & Kusano [1], which resolves five intermediate states separated by two fast magnetosonic waves (S_L, S_R), two rotational (Alfvén) discontinuities (S_{AL}, S_{AR}), and the contact/entropy wave (S_M). The fast wave speeds are estimated from the left and right mag-

netosonic speeds:

$$c_f^2 = \frac{1}{2}\left(a^2 + \frac{|\mathbf{B}|^2}{\rho} + \sqrt{\left(a^2 + \frac{|\mathbf{B}|^2}{\rho}\right)^2 - 4a^2\frac{B_x^2}{\rho}}\right), \tag{3}$$

where $a = \sqrt{\gamma p/\rho}$ is the sound speed. The contact wave speed S_M follows from pressure balance across the central region, and the double-star states are obtained by enforcing continuity of v_y, v_z, B_y, B_z across the Alfvén waves.

The HLLD solver is the single most expensive kernel in the code. I wrote it as a Numba JIT compiled loop over all interface faces, to give near-C performance while keeping the source in pure Python as my C skills are yet to catch up to this level.

C. Spatial Reconstruction: MUSCL–Van Leer

I use second-order MUSCL reconstruction with the Van Leer harmonic-mean slope limiter:

$$\phi(r) = \frac{2r}{1+r}, \quad r = \frac{\Delta_{i-1/2}}{\Delta_{i+1/2}}, \tag{4}$$

applied component wise in primitive variables. The Van Leer limiter is less diffusive than minmod while remaining TVD, which matters at contacts. Left and right reconstructed states $\mathbf{W}_{i+1/2}^{L,R}$ are passed directly to the HLLD solver at each cell interface.

D. Time Integration: SSP-RK3

Temporal integration uses the three stage, strong stabilitypreserving, Runge–Kutta scheme of Shu & Osher:

$$\mathbf{U}^{(1)} = \mathbf{U}^n + \Delta t\,\mathcal{L}(\mathbf{U}^n), \tag{5}$$

$$\mathbf{U}^{(2)} = \tfrac{3}{4}\mathbf{U}^n + \tfrac{1}{4}\big[\mathbf{U}^{(1)} + \Delta t\,\mathcal{L}(\mathbf{U}^{(1)})\big], \tag{6}$$

$$\mathbf{U}^{n+1} = \tfrac{1}{3}\mathbf{U}^n + \tfrac{2}{3}\big[\mathbf{U}^{(2)} + \Delta t\,\mathcal{L}(\mathbf{U}^{(2)})\big], \tag{7}$$

where $\mathcal{L}$ is the spatial operator (flux divergence plus source terms). The SSP property guarantees that if the forward-Euler update is TVD, so is the full RK3 step. The CFL number is fixed at 0.3.

E. Divergence Cleaning: GLM + Powell

The divergence constraint $\nabla \cdot \mathbf{B} = 0$ is not automatically satisfied by the discrete update. I enforce it using a two-pronged approach.

First, the Generalized Lagrange Multiplier method of Dedner et al. [2] couples an auxiliary scalar field ψ to the induction equation via a hyperbolic–parabolic subsystem:

$$\frac{\partial \psi}{\partial t} + c_h^2\,\nabla \cdot \mathbf{B} = -\frac{c_h^2}{c_p^2}\,\psi, \tag{8}$$

where c_h is the maximum signal speed and c_p controls damping. Divergence errors are advected out of the domain at speed c_h and simultaneously damped at rate c_h/c_p. After each RK stage, the scalar ψ is exponentially decayed:

$$\psi \leftarrow \psi\,\exp\!\left(-\alpha\,\frac{c_h\,\Delta t}{\min(\Delta x, \Delta y)}\right), \tag{9}$$

with $\alpha = 1.5$.

Second, I add the eight-wave Powell source terms [3] computed from a face-centered estimate of $\nabla \cdot \mathbf{B}$:

$$\mathbf{S}_{\text{Powell}} = -(\nabla\cdot\mathbf{B})\,\big(0,\, B_x,\, B_y,\, B_z,\, v_x,\, v_y,\, v_z,\, \mathbf{v}\cdot\mathbf{B},\, 0,\, 0\big)^T. \tag{10}$$

The face-centered differencing avoids the checkerboard modes that plague cell-centered divergence estimates. Together, GLM and Powell keep $|\nabla \cdot \mathbf{B}|$ well controlled throughout the simulation, as shown in the diagnostics.

III. VERIFICATION & VALIDATION

An MHD code that has not been verified against known solutions is generating pretty pictures, not physics. I ran three standard tests before touching the RMI problem.

A. Brio–Wu Shock Tube

The Brio–Wu problem [4] is a one-dimensional MHD Riemann problem that produces a compound wave structure: a fast rarefaction, a slow compound wave, a contact discontinuity, a slow shock, and a fast rarefaction, all interacting with a transverse field reversal ($B_y : +1 \to -1$). It is the canonical stress test for MHD Riemann solvers because any diffusion at the contact or mishandling of the compound wave is immediately visible.

I ran the test at $N_x = 800$ cells to $t = 0.1$ with $\gamma = 2$. The solver captures all five wave families cleanly: the density profile shows the correct plateau structure with sharp transitions, the B_y reversal is resolved within a few cells, and the velocity field matches the expected profile. No spurious oscillations appear at any discontinuity.

B. Linear Alfvén Wave Convergence

To measure formal order of accuracy, I propagated a small-amplitude (10^{-6}) circularly polarized Alfvén wave through a periodic domain for exactly one wave period. The L_1 error in B_y was measured at resolutions $N_x = 32, 64, 128, 256$.

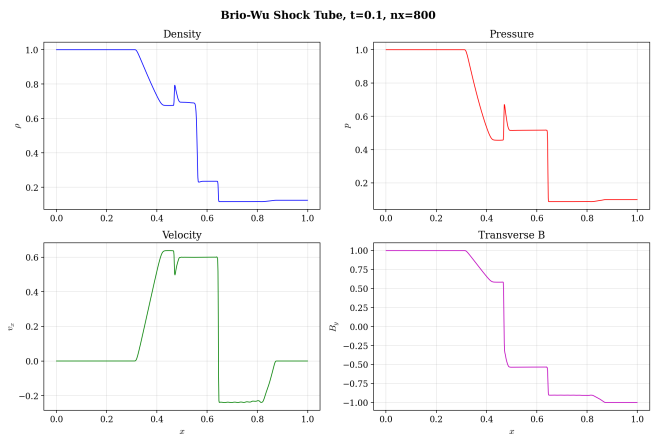

FIG. 1. Brio–Wu shock tube at $t = 0.1$, $N_x = 800$. The HLLD solver resolves all compound wave structures without oscillation.

The code recovers second-order convergence ($\mathcal{O}(\Delta x^2)$), consistent with the MUSCL reconstruction and SSP-RK3 integration. This test also validates the periodic boundary conditions and confirms that the GLM cleaning does not pollute a divergence-free eigenmode.

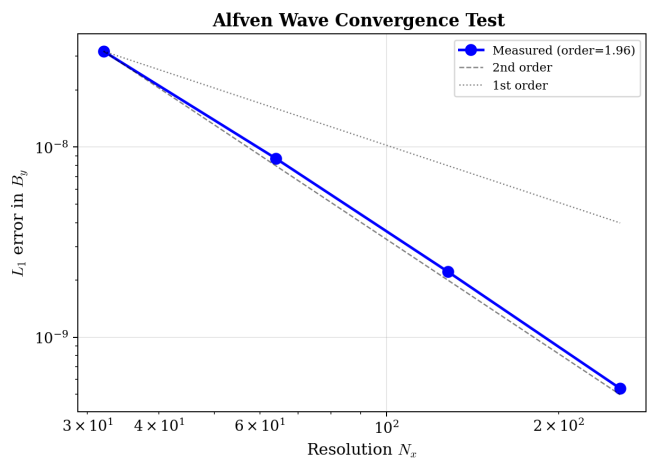

FIG. 2. Alfvén wave convergence test. The measured L_1 error follows the second-order reference slope.

C. Contact Discontinuity

An isolated contact discontinuity (density jump from 1 to 3, uniform pressure and velocity, $B_x = 0.5$) was advected across the domain. Pressure and velocity variations remained below 5% and 0.1 respectively, confirming that the HLLD solver does not generate spurious pressure fluctuations at contacts,a usual failure mode of HLL solvers that do not resolve the entropy wave as I have come to learn.

IV. PHYSICS RESULTS: MHD SUPPRESSION OF RMI

A. Problem Setup

A Mach 10 fast magnetosonic shock propagates in the $+x$ direction through a light gas ($\rho_1 = 1$, $p_1 = 1$) and impacts a perturbed density interface at $x = 1.5$. The heavy gas has density $\rho_h = 3$ (Atwood number $A \approx 0.5$ pre-shock). The interface carries a sinusoidal perturbation with mode number $k = 4$ and amplitude $a_0 = 0.15$ on a domain $[0, 6] \times [0, 2]$ at resolution 400×200.

I ran three cases: pure hydrodynamics ($B_y = 0$), moderate field ($B_y = 0.5$), and strong field ($B_y = 1.5$). The pre-shock Rankine–Hugoniot jump conditions, including the MHD corrections for transverse field compressio, are solved analytically and used to initialize the post-shock state. Characteristic, non-reflecting, boundary conditions are applied in x; periodic boundaries in y.

B. Morphology: Spikes, Bubbles, and Their Collapse

In the hydrodynamic case, the shock deposits baroclinic vorticity $\sim \nabla\rho \times \nabla p$ at the corrugated interface. The perturbation inverts phase (Richtmyer's prediction) and then grows into the familiar spike-and-bubble morphology. By $t = 0.25$, secondary Kelvin–Helmholtz roll-ups decorate the spike tips, and the mixing zone is broad and turbulent.

With $B_y = 0.5$, the large-scale structure survives but the secondary KH instabilities are visibly damped. The spike tips remain coherent and the mixing zone is narrower. At $B_y = 1.5$, the suppression is dramatic: the interface deforms but retains a nearly sinusoidal shape, the spike-and-bubble asymmetry is weak, and no secondary instabilities develop. The magnetic tension force $(\mathbf{B}\cdot\nabla)\mathbf{B}$ acts as a restoring spring on the interface corrugation.

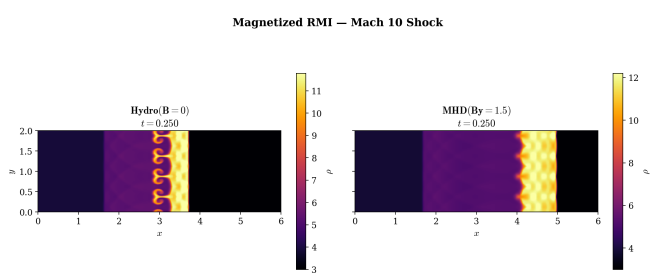

FIG. 3. Density fields at $t = 0.25$ for the hydrodynamic (left) and strong-field MHD (right) cases. Magnetic tension collapses the spike-and-bubble structure and preserves interfacial coherence.

C. Enstrophy and Mixing Width

I track two quantitative diagnostics. The integral mixing width is defined as

$$W(t) = \int \langle C \rangle_y \left(1 - \langle C \rangle_y\right) dx, \tag{11}$$

where $\langle C \rangle_y$ is the y-averaged passive scalar. This quantity peaks when the mixing zone is maximally interdiffused and drops as one species dominates any given column.

The local enstrophy, $\langle \rho \omega_z^2 \rangle$, is computed in a band around the interface and measures the intensity of rotational motion driving the mixing.

In the hydrodynamic case, both W and the enstrophy grow monotonically after shock passage. At $B_y = 0.5$, the mixing width is reduced and the enstrophy peaks lower. At $B_y = 1.5$, the enstrophy is suppressed by a large factor relative to the hydro baseline, and the mixing width remains small.

The mechanism is clear: Alfvén waves propagating along the field lines transport the baroclinic vorticity deposited at the interface away from the mixing zone. The vorticity is not destroyed, it is relocated. Simultaneously, the magnetic tension opposes the shear that drives KH roll-up, cutting off the secondary instability before it can broaden the mixing zone.

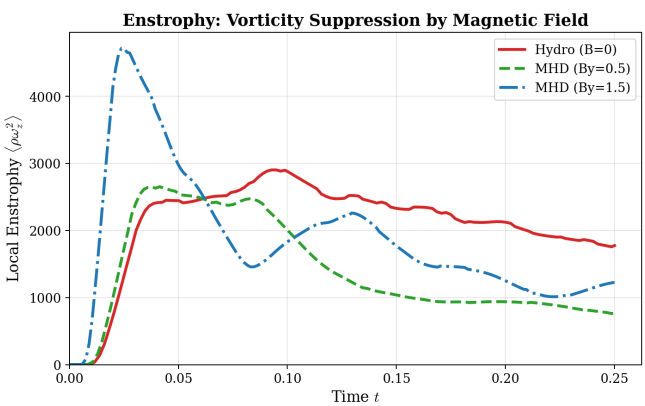

FIG. 4. Local enstrophy versus time for all three cases. The transverse field progressively suppresses vortical activity at the interface.

D. Linear Theory Comparison

In the early post-shock phase, the perturbation amplitude should follow Richtmyer's impulsive model:

$$\dot{a} = A^+ k \, a_0^+ \, \Delta v, \tag{12}$$

where A^+ is the post-shock Atwood number, $a_0^+ = a_0/r$ is the compressed amplitude (r is the density compression ratio), and $\Delta v = v_s(1 - 1/r)$ is the velocity impulse. For the MHD cases, I additionally account for the oscillatory behavior imposed by the Alfvén restoring force, replacing linear growth with

$$a(t) = a_0^+ \cos(\omega_A t') + \frac{\dot{a}_0}{\omega_A} \sin(\omega_A t'), \tag{13}$$

where $\omega_A = k \, v_A$, $v_A = B_y/\sqrt{\bar{\rho}}$, and $t' = t - t_{\text{shock}}$.

The simulation data matches the linear prediction well during the early phase. The hydrodynamic case tracks the linear growth rate before nonlinear saturation causes departure. The MHD cases show the expected oscillatory behavior, with the simulation following the predicted frequency and initial amplitude.

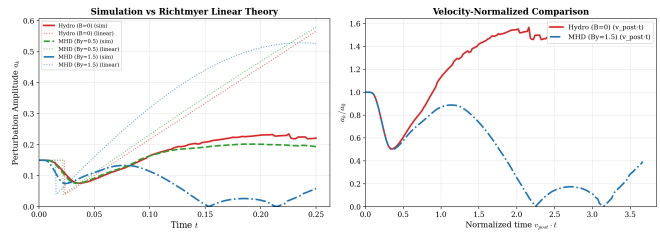

FIG. 5. Perturbation amplitude versus time compared to Richtmyer's impulsive linear model (dotted lines). Early-time agreement validates both the initial conditions and the shock-interface interaction.

V. CONCLUSION

The solver works. The HLLD scheme resolves all five MHD wave families, the MUSCL–Van Leer reconstruction delivers second-order accuracy, and the GLM–Powell cleaning keeps magnetic monopoles under control. Three independent verification tests confirm this before any physics is extracted.

The physics are equally clear. A transverse magnetic field of sufficient strength kills the Richtmyer–Meshkov instability at a Mach 10 shock. The suppression operates through two channels: Alfvén-wave transport of baroclinic vorticity away from the interface, and magnetic tension damping of secondary Kelvin–Helmholtz instabilities. The integral mixing width, local enstrophy, and perturbation amplitude all confirm the effect quantitatively, and the early-time growth rates agree with Richtmyer's impulsive linear theory.

The next step is obvious. This code is written in Python with Numba JIT, fast enough for 2D parameter sweeps, too slow for 3D production runs. The architecture, cell-centered finite volume, dimensionally split fluxes, explicit time stepping, maps directly onto GPU hardware. A CUDA or HIP port targeting a multi-GPU cluster would enable three-dimensional simulations at the resolutions needed to resolve the full turbulent cascade in MIF-relevant geometries.

The mixing zone is the problem. The magnetic field is the answer. The code proves it.

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